

Multi-Attach Amazon EBS volumes

To Begin with Lab

Summary of the Lab

To create a Windows EC2 instance on AWS, launch a new instance, select Microsoft Windows Server 2022 Base, and configure key pairs and security groups. Then, create a multi-attach volume (io1/io2) in the same availability zone, enable multi-attach, and attach it to the instance. Note: avoid using t2 AMIs for multi-attach volumes.

- Let's create an instance first.
- Go to AWS Console.
- Navigate to EC2.
- Click on launch an instance.
- Name you instance.
- In Application and OS Images (Amazon Machine Image), go to quick start
- Select Windows (Microsoft)
- In Amazon Machine Image (AMI), Select Microsoft Windows Server 2022 Base.

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name: my-vm-server Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or browse for AMIs if you don't see what you are looking for below.

Search our full catalog including 1000s of application and OS images

Recents My AMIs Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Microsoft Windows Server 2022 Base
ami-0345f44fe05216fc4 (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Summary

Number of instances: 1

Software Image (AMI)
Microsoft Windows Server 2022 ...read more
ami-0345f44fe05216fc4

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel Launch instance Preview code

Amazon Machine Image (AMI)

Microsoft Windows Server 2022 Base
ami-0345f44fe05216fc4 (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Microsoft Windows Server 2025 Base
ami-02b60b5095d1e5227 (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Microsoft Windows Server 2025 Core Base
ami-0909c0d1feca5b90c (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Microsoft Windows Server 2022 Base
ami-0345f44fe05216fc4 (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Microsoft Windows Server 2022 Core Base
ami-07c76fe41461bcf81 (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Microsoft Windows Server 2019 Base
ami-0ed9f06d3c0eb955a (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Microsoft Windows Server 2019 Core Base
ami-060c2b9960a8002fb (64-bit (x86))
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

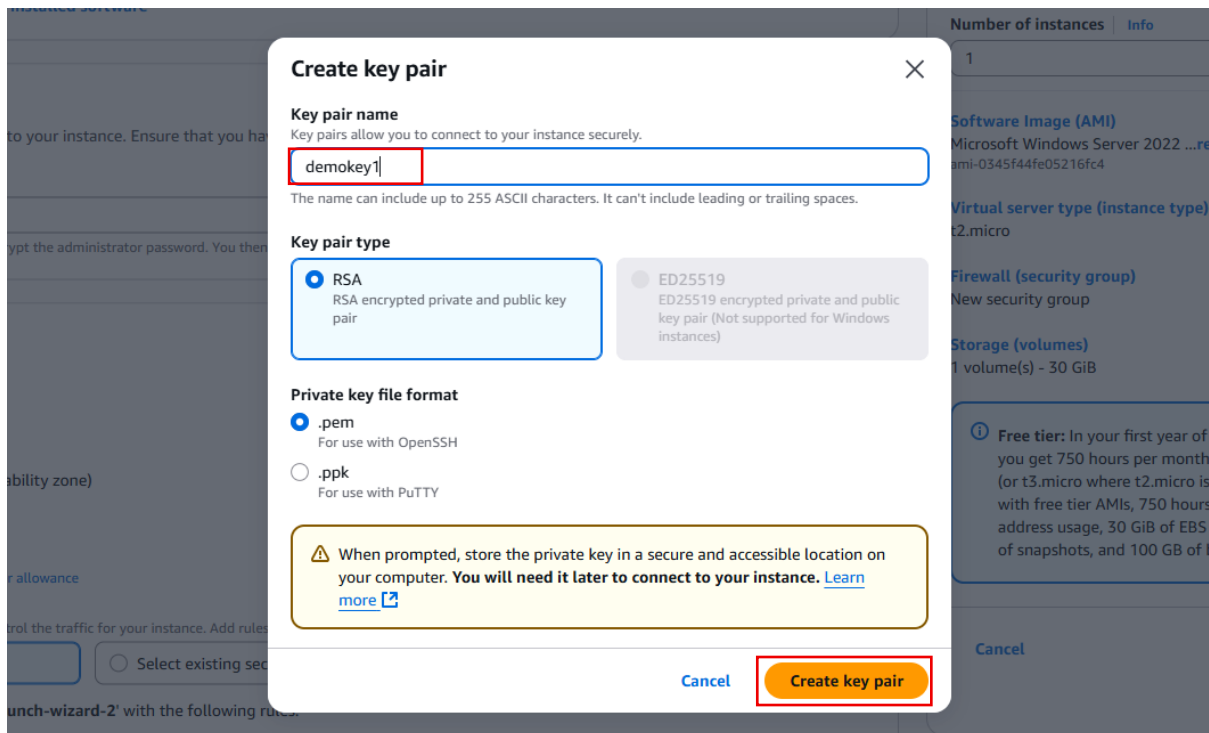
Storage (volumes)
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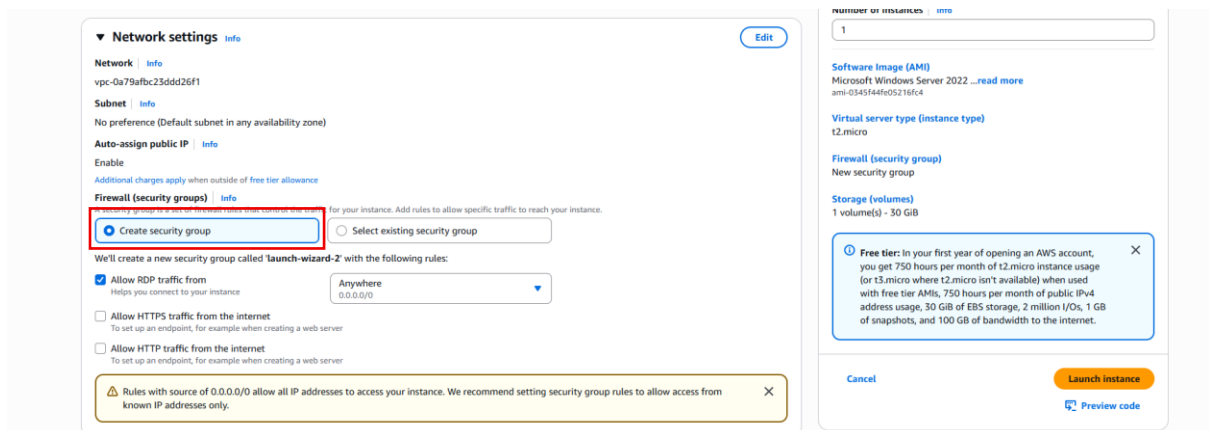
Cancel Launch instance Preview code

- Now in key pair (login)
- If you have existing key pair, you can use that.
- Otherwise, click on the 'Create new key pair'.

- Name your key pair.
- Click on the 'create key pair'
- Choose the options as given.



- It will download a '.pem' file that will be used to login to the virtual machine.
- Go to Network settings
- If you have an existing security group, select that.
- Otherwise, click on the create security group
- Choose the option given below.



- For now, I am not increasing the memory size for the root volume.
- You can increase according to your need.
- Click on the launch instance.

Configure storage [Info](#) Advanced

1x 30 GiB gp2 Root volume, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems [Edit](#)

[Advanced details](#) [Info](#)

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

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[Cancel](#) [Launch instance](#) [Preview code](#)

- Here is the new created instance.

Instances (1/2) [Info](#)

Last updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
myvm-server	i-0ed89c2f2ab0b36b1	Running	t2.micro	2/2 checks passed	View alarms	us-east-1b	ec2-3-83-90-126.comp...	3.83.90.126	-

Now let's create a multi-attach volume.

Go to Volume.

Create Volume.

Elastic Block Store

[Volumes](#)

[Snapshots](#)

[Lifecycle Manager](#)

Network & Security

[Security Groups](#)

Volumes [Info](#)

Last updated less than a minute ago

[Actions](#) [Create volume](#)

Saved filter sets
Choose filter set

Search

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created
You currently have no volumes in this region							

Choose provisioned IOPS SSD (io1) or (io2), size, IOPS, choose the same availability Zone in which your instance is created.

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)
Provisioned IOPS SSD (io1)

Size (GiB) [Info](#)
4
Min: 4 GiB, Max: 16384 GiB.

IOPS [Info](#)
100
Min: 100 IOPS, Max: 200 IOPS (up to 50 IOPS per GiB)

Throughput (MiB/s) [Info](#)
Not applicable

Availability Zone [Info](#)
us-east-1b

Check the Enable Multi-Attach

If you want to put encryption, you can check the Encrypt this volume.

Snapshot ID - optional [Info](#)
 Don't create volume from a snapshot ▼ 🔄

Amazon EBS Multi-Attach [Info](#)
☒ Enable Multi-Attach

🔔 Multi-Attach enabled volumes do not support I/O fencing. I/O fencing protocols control write access in a shared storage environment to maintain data consistency. Your applications must provide write ordering between instances to maintain data consistency on a Multi-Attach enabled volume.

Encryption [Info](#)
 Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.
☐ Encrypt this volume

Click create volume

Tags - optional [Info](#)
 A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
 No tags associated with the resource.
[Add tag](#)
 You can add 50 more tags.

Snapshot summary [Info](#) 🔄
 🔔 Click refresh to view backup information
 The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

[Cancel](#) [Create volume](#)

Now we will attach the volume to the instance.

Select the volume.

Click on the action > attach volume.

Volumes (1/2) [Info](#)

Saved filter sets ▼ [Choose filter set](#)

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created
☐		vol-0c79b4e4fcb918164	gp2	30 GiB	100	-	snap-0d49943...	2025/06/...
☒		vol-07415ef7380f1cdc6	io1	4 GiB	100	-	-	2025/06/...

Volume ID: vol-07415ef7380f1cdc6

Actions ⬆

- Modify volume
- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume**
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Manage tags
- Fault injection

Choose the instance you want to attach.

Choose the device name.

Click Attach volume

Note: your instance shouldn't be created in t2 AMI.

Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

 vol-07415ef7380f1cdc6

Availability Zone

us-east-1b

Instance [Info](#)

i-04851b60ee7763dc3
(test) (running)



Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)

xvdf

Recommended device names for Windows: /dev/sda1 for root volume. xvdf[p-j] for data volumes.

[Cancel](#)

[Attach volume](#)

And your volume will be attached.